Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

Disaster Management System:A digital rescue shield

Team Name: *Team #*	*DSCPP-III-2026-T058* *ShieldNova*
Team member 1 (Team Lead)	*Mittal,Avika – 2510012768* *avikamittal123@gmail.com* 
Team member 2	*Singh Negi,Gaurav – 2510012516* *gauravnegi62007@gmail.com* 
Team member 3	*Khandelwal,Avni – 25103140050* *avnik1921@gmail.com* 
Team member 4	*Gautam,Aryan – 2510012654* *gautamaryan195@gmail.com* 

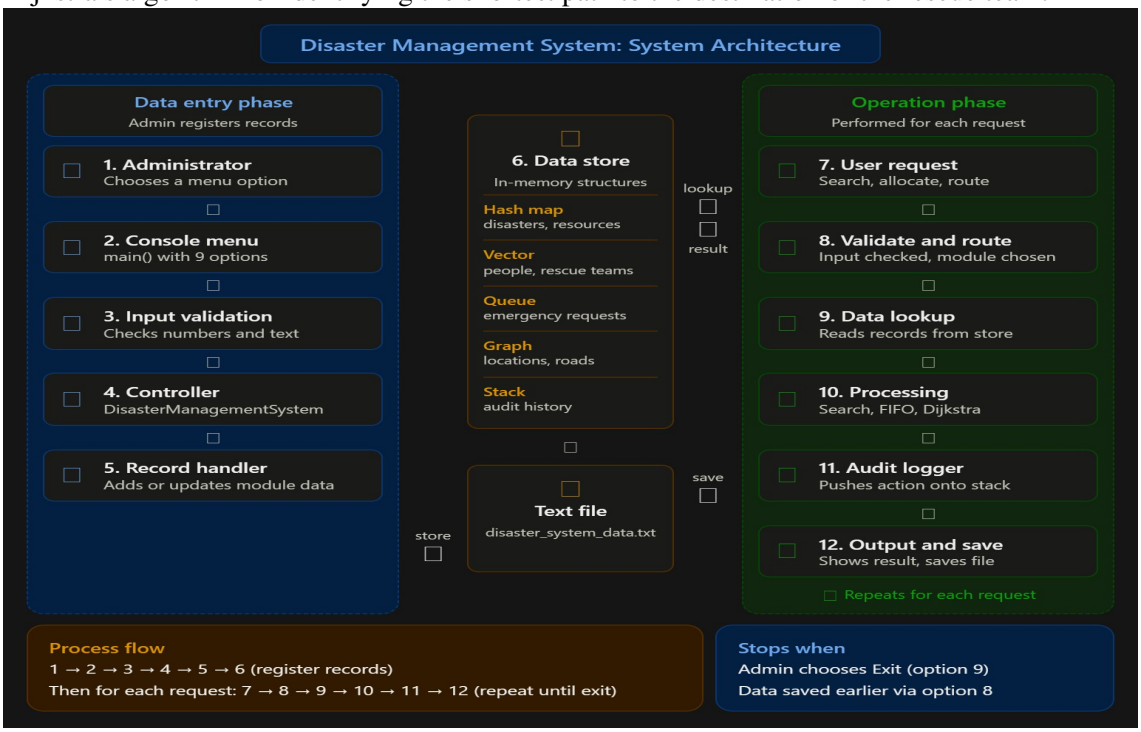
PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

A C++-based project called the Disaster Management System was created to efficiently and systematically arrange and manage information pertaining to disasters.The task of managing chaotic and disorganized data during the disaster management process becomes a complete mess as far as tracing victims,affected areas,rescue teams,and inventory is concerned.That's precisely why I developed a console application in C++ to efficiently manage all the data storage,searches,updates,and processing tasks.As far as its working is concerned,the entire project is based on Object Oriented Programming techniques along with traditional Data Structures and Algorithms.For this,I have made use of arrays,linked lists,stacks,queues,and graphs,which were complemented by traditional sorting and searching methods to handle different data record management in different modules of locations,resources,and disaster log information.Some of the unique aspects of my application are that it comes with a dashboard for the severity of the situation and automated dispatching of rescue teams.Dijkstra's algorithm was used to find the shortest possible path of rescue.The project includes file handling functions to prevent any loss of information during shutdowns of the program.

Updated Project Approach and Architecture

This software program created to be run on the console has been developed in Visual Studio Code/Code::Blocks and begins with systematic problem identification,need analysis,and process design in order to develop an efficient disaster management system.Arrays and linked lists are utilized by the software in managing disaster records,victims,rescue teams,and inventory without any hassle.In order to cope with stressful scenarios effectively,I have used priority queues for emergency dispatches and graph structures using Dijkstra's algorithm for identifying the shortest path to the destination of the rescue team.



Tasks Completed

Task Completed	Team Member
Requirements analysis and goal definition for the project.Initial workflow and system design.	Avika Mittal
Module for Basic Disaster Management - Development.Methodology,Development Strategy and Initial Project Documentation.	Gaurav Singh Negi
Selection of suitable DSA concepts for system functions.Route finding,resource allocation,rescue routing,and severity dashboard concept development.	Avni Khandelwal
Disaster databases and sample data creation.	Aryan Gautam

Challenges/Roadblocks

1.Integration of Data Structures: Trying to cram arrays,linked lists,stacks,queues,and graphs all into one app can get pretty messy and complicated.To keep things from breaking,our plan is to build and test each data structure completely on its own before we try gluing the modules together.
2.Emergency Prioritization: Sorting out disaster cases by how severe they are takes some careful logic.We're setting up a priority queue along with proper sorting methods,and we'll test it out thoroughly with a bunch of different severity levels to make sure it works right.
3.Graph-Based Route Determination: In order to calculate the shortest escape route by running the Dijkstra algorithm,precise calculation of the distance and sound graph configuration is essential.In order to make sure that the logic of the algorithm is true,we will first create some test data points.
4.File Handling: Reading text files could become problematic if the formats used for data storage got corrupted and files could not be saved or loaded during disaster record management. In order to prevent any data loss,we will have to be particularly cautious about using C++ file streams and checking all read/write operations.

Tasks Pending

Task Pending	Team Member (to complete the task)
1.Make sure that there is uniformity in the flow of data across the system.	Avika Mittal
2.Use linked lists and queues to allocate rescuers. comprehensive resource management,which involves resource allocation,up-to-date and search capabilities.	Gaurav Singh Negi
3.Use of priority queue and sorting technique in prioritizing emergencies. Use of Dijkstra's algorithm in finding the rescue route using graphs.	Avni Khandelwal
4.Disaster records can be saved and retrieved through file handling. System Testing, Debugging, Validation, and Documentation.	Aryan Gautam

Project Outcome/Deliverables

1.Key Project Outcomes
The primary goal of the Disaster Management System is to create a C++ application that arranges data related to disasters and shows how Data Structures and Algorithms (DSA) and Object-Oriented Programming (OOP) can be used practically.The system's goal is to make managing disaster records,impacted individuals,rescue teams,emergency requests,resources,and locations easier.By using appropriate data structures,sorting,and searching methods,it is anticipated to enhance information organization and retrieval.
2.Project Delierables
The following are the project's primary deliverables:
An application for managing records related to disasters that is modular in C++. Dashboard for managing the severity of disaster that categorizes emergencies in terms of severity. Modules for resources management and for rescuer dispatching. Algorithm of Dijkstra to determine optimal rescue routes using graphs.Text files are used to save and retrieve record of data because of file handling capabilities.
Documentation for the project, test scenarios, sample data sets, and source code.Concluding demo explaining how the system works along with explaining concepts of DSA and OOP used.The above deliverables should demonstrate how effective disaster data management can be accomplished using fundamental concepts of computer science. After implementing and integrating all proposed components, the entire system is validated.

Progress Overview

1.Status of Project Completion
The first phase of the project's planning and design has been finished.The system architecture,workflow,problem statement,project objectives,and necessary data structures and algorithms have all been determined.Sample disaster data has been prepared,and the basic disaster management module has been developed.Additionally described are the designs for resource management,graph-based route finding,emergency prioritization,and rescue team dispatch.Completing these features,integrating the modules,and testing the entire application comprise the majority of the remaining work.
2.Status of the Schedule
So far we have been able to complete the initial architecture design,requirement analysis,and DSA selection within schedule. However,the other areas such as file handling,proper testing,and coding of advanced features remain to be done and can consume some more time.Nothing so far seems to have gone ahead of schedule. In order to ensure that the entire process is completed before time,our strategy will be to work on the outstanding modules first and conduct proper testing.

Codebase Information

1.Branch and Repository Link
We're keeping all our source code in a GitHub repository so version control and team collaboration go smoothly.As soon as the repo is created and verified,I'll update it with the actual link.We're planning on using main as the default branch,though we'll want to double-check the exact branch name once everything is set up. C++ is the programming language.The development environment Code::Blocks in Visual Studio Code.
2.Important Commits
Git commits will be used by the team to monitor the creation and modification of project modules.The following are anticipated to be significant commits:
Preliminary requirements documentation and project structure.
Implementation of a basic disaster management module.
Addition of sample data and disaster records.
Putting emergency resource management and prioritization into practice.
Graph-based route finding and rescue team dispatch are integrated.
File management,testing,debugging,and integration of the final code.
As development moves forward,commit dates and details will be added based on the real Git history.

Testing and Validation Status

Test Type	**Status (Pass/Fail)**	**Notes**
Disaster Record Operation	Implemented	Examine how to register,update,view,and deal with missing IDs.
Affected People Management	Implemented	Try adding,looking up,and changing a person's status.
Rescue Team Management	Implemented	Test the registration process and show the team's availability.
Resource Allocation	Implemented	Check FIFO behavior and processing when the queue is empty.

Deliverables Progress

The primary modules for disaster management,affected people management,rescue team management,resource allocation,emergency request processing using a queue,shortest-path routing using Dijkstra's algorithm,and action history using a stack have all been completed for the Disaster Management System,along with its fundamental menu-driven interface.Right now,file handling is only half-done—I can save the records for affected individuals,but loading them back up or saving data from the other modules is still a work in progress.On top of that,emergency requests are just running on a standard FIFO queue right now,so I still need to implement priority-based scheduling so the most urgent calls get handled first.
Beyond writing the core code,I've still got to tackle thorough testing,bug fixes,final checks,putting together proper documentation,and getting the final demo ready.These notes are just based on where the code stands at this exact moment,and I definitely need to run some solid tests to make sure everything is working the way it should.